

1.3 Primes and Unique Factorization

Goal: We want to show that every integer is a product of primes, that this factorization is unique in some sense.

Example. Find a prime factorization of 12.

Definition. An integer p is prime if $p \notin \{0, \pm 1\}$, and the only divisors of p are ± 1 and $\pm p$.

Remark: In many other sources, prime numbers are defined to solely include positive integers. However we will stick to the textbook's convention.

Examples of primes

Facts about primes and divisibility.

- There are infinitely many distinct primes (Group Exercise) .
- Let p and q be prime integers. Then $p \mid q$ if and only if $p = \pm q$.

Theorem 1.5. Let $b, c \in \mathbb{Z}$, and let p be an integer with $p \notin \{0, \pm 1\}$. Then p is prime if and only if p satisfies the condition:

$$\text{if } p \mid bc, \text{ then } p \mid b \text{ or } p \mid c.$$

Recall Theorem 1.4, If $a \mid bc$ **and** $\gcd(a, b) = 1$, then $a \mid c$.

Proof. $(\Rightarrow)$:

$(\Leftarrow)$:

Remark. Because Theorem 1.5 is biconditional, we've just given a **new** definition of what it means to be prime.

Corollary 1.6. If $a_1, a_2, \dots, a_n \in \mathbb{Z}$, p is prime, and $p \mid a_1 a_2 \cdots a_n$, then p divides at least one of the a_i .

Proof. Sketch:

Theorem 1.7. Every integer $n \notin \{0, \pm 1\}$ is a product of primes.

Proof.

Remark: An integer other than $0, \pm 1$ that is not prime is called a composite

Remark. This theorem gives the *existence* of a prime factorization for every integer $n \neq 0, \pm 1$. Theorem 1.8 gives the (almost) uniqueness of this factorization.

Theorem 1.8.[Fundamental Theorem of Arithmetic](FTA) Every integer $n \notin \{0, \pm 1\}$ is a product of primes, and these primes are unique in the following sense:
If $n = p_1 p_2 \dots p_r$ and $n = q_1 \dots q_s$ for some primes p_i, q_j , then $r = s$. Also for all $i \in \{1, \dots, r\}$ we have $p_i = \pm q_j$ for some $j \in \{1, \dots, r\}$

Proof Strategy: We will use Mathematical Induction.

I **Base Case:** First we show that FTA holds when $r = 2$.

II **Inductive Step:** Next we assume FTA holds for $r - 1$ and show that this implies FTA for r

Notice then that by I the theorem holds for $r = 2$. By II the theorem holds for $r = 3$. By II the theorem holds for $r = 4$. By II the theorem holds for $r = 5$ and so on. Therefore FTA will hold for all r .

Proof.

Group Exercises.

1. (Theorem 1.10) Prove, by contradiction, that if n is an integer greater than 1 and n is not prime, then n is divisible by some prime $p \leq \sqrt{n}$.
2. Show that $\sqrt{2}$ is irrational using the Fundamental Theorem of Arithmetic.
3. Show there exists an infinite number of primes. (Hint: Try doing this by contradiction.)